Animation Handoff Pack

Everything needed to animate this game without opening it: the four rigs and what is wrong with each, the scene and export spec, the animator contract, and the exact frame budget for every move. All figures are measured from the project as it stands on 21 October 2026.

SCOPE

Fighters	4
Animator states	90
Attacks with frame data	33
Clips authored for this game	10
States on borrowed pack clips	80
States playing faster than 1.8x	37

READ THIS FIRST

The game is animated almost entirely from three bought Mixamo packs. Where a pack clip is longer than the move it was assigned to, the import tooling **speeds it up to fit** — and 37 of 90 states are being played at 1.8x or more. That, not the rigging, is what makes the game look clumsy.

Every move table in here gives a **Target** length and a **@30fps** frame count. An animation keyed to that number plays at x1.00 and the problem disappears.

The four rigs

Two different auto-riggers are in play, and they do not share a bone naming convention. Nothing here was rigged by a person except the corrections on Salaadin.

FIGHTER	RIG	BONE NAMES	BONES	NODES	VERTS	HEIGHT	SCENE X	STATE
Bjorn	Tripo auto-rig	`R_Hand, …`	~40	45	144,779	1.01	4.0	OK
Miyamoto	Tripo auto-rig	`R_Hand, …`	~40	45	139,431	1.04	3.0	OK
Richard	Mixamo auto-rig	`mixamorig:*`	65	70	14,649	2.04	3.0	OK
Salaadin	Mixamo, **hand-corrected**	`mixamorig:*`	65	70	12,574	1.09	3.0	FIXED

Height is the skinned bounds in model units. **Scene x** is the uniform scale on the fighter root in the arena — the models are different sizes, so the scene corrects for it. Gameplay distances are authored in world units and never derived from the model scale.

WEAPON MOUNTS

FIGHTER	RIGHT HAND	LEFT HAND	MOUNT NOTES
Bjorn	`Bjorn_Axe_R`	—	parented to `R_Hand`; two-handed grip constraint
Miyamoto	`Miyamoto_Katana_R`	—	parented to `R_Hand`; two-handed grip constraint
Richard	`Richard_Sword_R`	`Richard_Shield_L`	shield boss on `+palm`; no two-hand grip
Salaadin	`Salaadin_Falchion_R`	`Salaadin_Shield_L`	shield boss on `-palm`; falchion mounted with a -20 deg blade lean

Weapons are rigid children of the hand bones — **not skinned, and never keyed**. They follow the hand and nothing else, so a grip change has to be animated in the wrist.

WHAT IS WRONG WITH SALAADIN'S RIG, AND WHAT WAS DONE

His mesh is mostly cloth: leg geometry stops at the top of the boots and above it there is only coat skirt and cloak. Mixamo fits its skeleton to mesh *volume*, so where the thigh and pelvis should be it found only fabric. Two marker passes were tried; the second came back symmetric and worse. Four joints are now placed **by construction** in `salaadin_fixrig.py`, which moves bones in edit mode — the rest pose changes while the mesh and vertex groups stay put, so Mixamo's weights are kept and nothing is re-weighted.

JOINT	AS MIXAMO LEFT IT	CORRECTED TO	SYMPTOM IT CAUSED
Elbow	35% shoulder to wrist	52%	upper arm shorter than the forearm — the "snake arm"
Knee	bent 128 deg into the skirt	179 deg, on the hip-ankle line	a permanent crouch; never a clean T-pose
Hip separation	19.8% of height	11.0%, at 52% up	thighs converging twice as hard as the ankles
Hip fore/aft	forward, inside the coat	over the ankle	**the splits** — thigh rest tilted 23 deg backward

Bone roll is preserved throughout — a roll angle describes different axes once the head or tail has moved. **His arms never showed any of this**, because the generated clips key arms with IK and only the legs are copied rest-relative.

Scene and export spec

AUTHOR IN		DELIVER TO	
Application	Blender 5.1	**Destination**	`Animation/Clips/Custom/`
Scene frame rate	**30 fps**	**File name**	`<Char>_<Slot>_<Move>.fbx`
Game frame rate	60 fps (frame data)	**Contents**	armature + one empty, no mesh
Conversion	1 game frame = half a Blender frame	**The empty**	`<Char>_ClipRoot`
Up / forward	Y up, -Z forward (on export)	**Root motion**	**none** — see below
First keyed frame	**1** — frame 0 is reserved	**Scale**	FBX_SCALE_NONE, unit scale applied

No root motion. All movement is code.

`applyRootMotion` is **false** on all four fighters. Walking, dash distance, attack lunge and knockback are driven by the character scripts in world units. A clip that travels will be ignored at best and will fight the code at worst.

Animate in place. If a move *reads* as a lunge — Shield Charge, Veiled Rush — the travel is already in the game's lunge values; the clip supplies the lean and the footwork only.

FOUR IMPORT RULES EVERY AUTHORED CLIP MUST SATISFY

RULE	WHAT BREAKS WITHOUT IT
Copy the body's avatar CopyFromOther	Retargeting through the clip's own auto-built avatar is not an identity even on the same skeleton. Wrist twist does not survive it.
Based Upon: Original	*Body Orientation* infers facing from the posed body. On a bladed guard it read the turned shoulders as forward and yawed the whole skeleton **66.6 deg**. The tell is that every bone is off by the same angle — check that before blaming a wrist or a pole vector.
Export an empty beside the armature	An armature-only FBX makes the armature the file root, so Blender's 100x lands on the avatar's root. The two avatars then disagree on scale and every stretched joint is silently rewritten — the guard arrived with the elbows 16 deg tighter.
Frame 0 holds the rest pose	Blender writes the first exported frame as the file's bind pose, and Unity builds the clip's avatar from it. A clip that opens in the guard makes *the guard* the T-pose, and every frame returns as a delta from the wrong place.

All four are handled automatically by `EXPORT_CLIP` inside `Salaadin_Studio.blend` and by the `*_clips.py` generators. They are listed because any other export route has to reproduce them.

THE ROUND TRIP

1 · Key
Pose the rig in Blender from frame 1.

→

2 · Export
Run `EXPORT_CLIP`; the FBX lands in `Clips/Custom/`.

→

3 · Import
Unity: `Crucible > Animation > Use Mixamo packs`.

→

4 · Wire
Point the animator state at the clip; set speed back to x1.00.

The trap that costs a day: Unity caches the skeleton

`humanDescription.skeleton` stores every bone's position in the `.meta`, and Unity builds the avatar from that **cache, not the file**. Re-export an FBX with a joint in a new place and nothing changes at runtime. The tell is that the same hierarchy measures **correctly in edit mode and wrong in play mode**. `Rebuild Player<N>` clears the cached arrays before every reimport; anything else that touches a rig must do the same.

The animator contract

Each fighter has his own controller (`P1_Bjorn` … `P4_Salaadin`) but they share one parameter set and one skeleton of states. A new clip replaces the motion on an existing state; it does not add one.

PARAMETERS THE CODE DRIVES

PARAMETER	MEANING
`MoveSpeed`	float, walk blend, every frame
`LightAttack`	trigger
`MediumAttack`	trigger
`HeavyAttack`	trigger
`HeavyFromLight`	trigger, combo link
`SpecialLight`	trigger
`SpecialMedium`	trigger
`SpecialHeavy`	trigger
`RiposteDash`	trigger, dodge counter
`RiposteParry`	trigger, parry counter
`GetHit`	trigger
`HitSide`	int: 0 front, 1 left, 2 right
`Blocking`	bool
`Dash`	trigger
`DashDir`	int: fwd / back / left / right
`ParrySuccess`	trigger
`Death`	trigger

STATES EVERY FIGHTER HAS

STATE	LOOP	DRIVEN BY
`<Name>_Idle`	**yes**	default
`<Name>_Walk_Forward`	**yes**	MoveSpeed
`<Name>_Walk_Back`	**yes**	MoveSpeed
`Block_Hold`	**yes**	Blocking
`Block_Impact`	no	blocked hit
`React_Front`	no	HitSide 0
`React_Left`	no	HitSide 1
`React_Right`	no	HitSide 2
`Dash_Forward`	no	DashDir
`Dash_Back`	no	DashDir
`Dash_Left`	no	DashDir
`Dash_Right`	no	DashDir
`Parry_Success`	no	ParrySuccess
`Death`	no	Death

Plus the character's own 3 normals, 3 specials, 2 ripostes and any combo-link states. **Four clips per fighter must loop seamlessly**: idle, both walks, block hold.

HOW AN ATTACK IS TIMED

Every attack is three counts at 60 fps. The hitbox arms the instant startup ends and disarms when the active window closes; one hit per swing. **The contact frame is therefore `startup + 1`** — that is where the blade must be on the opponent, and the tables that follow give it for every move.



Active windows are 3 to 5 frames across the whole roster — 50 to 83 ms. The swing must read as a hit in that window, so the blade should be travelling fastest there, not arriving there.

> **Target length, and why it is the number that matters**
>
> Nothing stretches a clip automatically. `Use Mixamo packs` computes each state's speed itself — `state.speed = clip length / move length` — so a clip longer than its move is simply played faster, and **the speed is recomputed every time that menu item runs**. Editing a state's speed by hand will not survive.
>
> The **Target** column in the move tables is that divisor: `total frames / 60`, the length a clip has to be for the computed speed to come out at x1.00. The **@30fps** column is the same number in Blender frames, which is what to key to.
>
> A move with 25 total frames wants a **0.42 s clip, which is 13 Blender frames**. That is short. Fighting-game animation is short; the readability comes from the pose at contact, not from the number of frames around it.

Move specs — Bjorn and Miyamoto

Frames are 60 fps. **Contact** is the first active frame. **Target** is the clip length that plays at x1.00; **@30fps** is that in Blender frames. **Now** is the speed the current clip is being played at — shaded rows are stretched past 1.4x.

BJORN

MOVE	TRIGGER	START	ACT	REC	TOTAL	CONTACT	TARGET	@30FPS	SOURCE	NOW
Slash	`LightAttack`	10	4	8	**22**	11	0.37s	**11**	AxePack	×1.80
Spinning Slash	`MediumAttack`	18	5	12	**35**	19	0.58s	**18**	AxePack2H	×1.80
Overhead Double Axe	`HeavyAttack`	18	5	15	**38**	19	0.63s	**19**	GreatSwordPack	×1.80
Spinning Overhead	`HeavyFromLight`	28	4	14	**46**	29	0.77s	**23**	GreatSwordPack	×2.00
Blood Eagle	`SpecialHeavy`	24	5	24	**53**	25	0.88s	**26**	AxePack	×2.20
Haft Strike	`SpecialLight`	18	5	20	**43**	19	0.72s	**22**	AxePack	×2.00
Hewing Charge	`SpecialMedium`	22	5	22	**49**	23	0.82s	**24**	AxePack	×2.00
Axe Riposte	`RiposteDash`	12	4	14	**30**	13	0.50s	**15**	AxePack	×2.00
Boot Riposte	`RiposteParry`	12	4	14	**30**	13	0.50s	**15**	AxePack	×1.60

MIYAMOTO

MOVE	TRIGGER	START	ACT	REC	TOTAL	CONTACT	TARGET	@30FPS	SOURCE	NOW
Mid Thrust	`LightAttack`	6	3	6	**15**	7	0.25s	**8**	GreatSwordPack	×1.80
Side Slash	`MediumAttack`	9	4	8	**21**	10	0.35s	**10**	GreatSwordPack	×1.80
Overhead Slash	`HeavyAttack`	16	4	12	**32**	17	0.53s	**16**	GreatSwordPack	×2.00
Iaijutsu	`SpecialHeavy`	30	4	8	**42**	31	0.70s	**21**	GreatSwordPack	×1.25
Nito Cross	`SpecialMedium`	20	5	22	**47**	21	0.78s	**24**	GreatSwordPack	×2.26
Nukitsuke	`SpecialLight`	18	5	20	**43**	19	0.72s	**22**	GreatSwordPack	×1.00
Spinning Riposte	`RiposteDash`	12	4	14	**30**	13	0.50s	**15**	GreatSwordPack	×1.20
Push Kick Riposte	`RiposteParry`	20	4	6	**30**	21	0.50s	**15**	GreatSwordPack	×1.43

Move specs — Richard and Salaadin

RICHARD

MOVE	TRIGGER	START	ACT	REC	TOTAL	CONTACT	TARGET	@30FPS	SOURCE	NOW
Lion Thrust	`LightAttack`	11	3	9	**23**	12	0.38s	**12**	**authored**	×1.21
Cross Slash	`MediumAttack`	15	4	11	**30**	16	0.50s	**15**	SwordShieldPack	×1.60
Crusader Hew	`HeavyAttack`	22	5	17	**44**	23	0.73s	**22**	SwordShieldPack	×1.40
Twin Hew	`SpecialHeavy`	30	5	20	**55**	31	0.92s	**28**	**authored**	×2.36
Shield Bash	`SpecialMedium`	22	5	20	**47**	23	0.78s	**24**	**authored**	×2.16
Shield Charge	`SpecialLight`	24	5	18	**47**	25	0.78s	**24**	**authored**	×1.16
Lion Riposte	`RiposteDash`	12	4	14	**30**	13	0.50s	**15**	SwordShieldPack	×1.00
Boot Riposte	`RiposteParry`	12	4	14	**30**	13	0.50s	**15**	SwordShieldPack	×1.20

SALAADIN

MOVE	TRIGGER	START	ACT	REC	TOTAL	CONTACT	TARGET	@30FPS	SOURCE	NOW
Saif Cut	`LightAttack`	8	3	7	**18**	9	0.30s	**9**	SwordShieldPack	×1.11
Shin Cut	`MediumAttack`	12	4	9	**25**	13	0.42s	**12**	**authored**	×2.72
Whirling Saif	`HeavyAttack`	17	4	13	**34**	18	0.57s	**17**	**authored**	×2.71
Crescent Slash	`SpecialMedium`	20	5	19	**44**	21	0.73s	**22**	SwordShieldPack	×1.77
Hattin Cleave	`SpecialHeavy`	26	5	20	**51**	27	0.85s	**26**	SwordShieldPack	×2.71
Veiled Rush	`SpecialLight`	20	5	18	**43**	21	0.72s	**22**	**authored**	×2.00
Falcon Riposte	`RiposteDash`	12	4	14	**30**	13	0.50s	**15**	SwordShieldPack	×3.47
Saber Riposte	`RiposteParry`	12	4	14	**30**	13	0.50s	**15**	**authored**	×1.47

> **Salaadin fights with the falchion and nothing else**
>
> No thrusts, no shield bashes, no kicks. Where the other three answer a parry with a boot, his riposte is a saber counter. The shield blocks; it never strikes. Keep that when adding moves.

What is real and what is placeholder

Mixamo pack clip Authored for this game

FIGHTER	ALL ANIMATOR STATES	PACKS DRAWN ON	STRETCHED 1.8X+
Bjorn	 23 pack · 1 authored · 24 states	AxePack 12, AxePack2H 8, GreatSword 3	**16**
Miyamoto	 22 pack · 0 authored · 22 states	GreatSword 13, AxePack2H 7, AxePack 2	**12**
Richard	 18 pack · 4 authored · 22 states	SwordShieldPack 18	**3**
Salaadin	 17 pack · 5 authored · 22 states	SwordShieldPack 17	**6**

THE WORST STRETCHES IN THE GAME — THE WORK QUEUE

FIGHTER	STATE	CLIP	SOURCE	PLAYED AT
Bjorn	`React_Right`	1.70s	AxePack	**×3.78**
Salaadin	`Salaadin_Riposte_Dash`	1.73s	SwordShieldPack	**×3.47**
Miyamoto	`Parry_Success`	1.33s	AxePack	**×3.33**
Bjorn	`React_Front`	1.50s	AxePack	**×3.33**
Bjorn	`Parry_Success`	1.33s	AxePack	**×3.33**
Miyamoto	`Dash_Back`	0.68s	AxePack2H	**×3.11**
Miyamoto	`Dash_Forward`	0.67s	AxePack2H	**×3.03**
Miyamoto	`Dash_Right`	0.63s	AxePack2H	**×2.88**
Miyamoto	`React_Right`	1.27s	GreatSwordPack	**×2.81**
Miyamoto	`Dash_Left`	0.62s	AxePack2H	**×2.80**
Salaadin	`Salaadin_Medium_ShinCut`	1.13s	authored	**×2.72**
Salaadin	`Salaadin_Special_HattinCleave`	2.30s	SwordShieldPack	**×2.71**
Salaadin	`Salaadin_Heavy_WhirlingSaif`	1.53s	authored	**×2.71**
Miyamoto	`React_Left`	1.20s	GreatSwordPack	**×2.67**
Richard	`Richard_Special_DoubleHigh`	2.17s	authored	**×2.36**
Bjorn	`Dash_Back`	0.68s	AxePack2H	**×2.28**

Top 16 of **37 states at 1.8x or worse**, out of 90. Bjorn and Miyamoto hold 28 of the 37 — the Axe and Great Sword packs are long cinematic clips pinned to short moves. Richard is cleanest at 3, because the Sword and Shield pack happens to be brisk.

Where to start, and what to watch for

SUGGESTED ORDER

#	WORK	WHY IT IS FIRST
1	**The three reaction clips, per fighter**	React_Front / Left / Right run up to x3.78. They fire on *every single hit*, so they are the most-seen animation in the game and the worst-played.
2	**Dashes, Bjorn and Miyamoto**	x2.3 to x3.1, and dashes are the defensive core — they fire constantly and read as a twitch.
3	**Salaadin's Medium and Heavy**	x2.72 and x2.71. They are his two showpiece normals and already authored, so only the timing needs redoing.
4	**Parry_Success and the ripostes**	x3.33 and x3.47. The payoff beat of the whole defensive game lands as a blur.
5	**Specials**	x1.8 to x2.4. Most visible per use, least frequent — real value, but after the above.

THINGS THAT WILL BITE

WATCH FOR	DETAIL
Two bone conventions	Bjorn and Miyamoto are Tripo rigs with their own names; Richard and Salaadin are `mixamorig:*`. A clip authored on one will not open on the other. Pack clips cross between them only because they go through Unity's humanoid retargeting.
Porting a pose between rigs	Convert, never scale by height. Limb targets belong in **fractions of that limb's own reach**; height-scaled metres miss. Swing a limb with two-bone IK, not by rotating a bone about its own axis — bone roll differs per rig and the limb comes out sideways.
Elbow poles	The pole must sit **below** the shoulder. Salaadin's was once written level with it and both arms came out winged.
Salaadin's stance	`LEG_DAMP = 0.55` in the generator eases his legs back toward rest, because the pack's block idle plants the feet 94 deg apart — fine on Richard, the splits on him. Keying legs by hand sidesteps this entirely; do not copy the generated leg values.
Bodies are not in git	The four body FBXs and all source meshes are gitignored and live on one machine. They are rebuilt from the Blender scripts. The six weapon FBXs *are* committed.
The frame data can move	Startup / active / recovery live in one table, `Editor/CrucibleBalance.cs`. Change a move's timing there and its clip is re-fitted the next time `Use Mixamo packs` runs — so a balance pass silently re-times animation. Re-run that menu item after any frame change, then check the speed.

> **The file to work in**
>
> `ArtSource/Blender/Salaadin_Studio.blend` opens on his corrected rig in pose mode, falchion and shield mounted, mesh locked, scene at 30 fps. Two text blocks ride inside it: **EXPORT_CLIP** sends the current action to Unity with all four import rules satisfied, and **README** repeats the workflow. The export runs on a throwaway copy, so it never touches the animation being worked on.
>
> The equivalent for Richard is the generator `richard_clips.py`; a studio file for him has not been built yet.